

Minimizing Contractor Reliance & Credit Purchases to Strengthen Direct Customer Relationships and Improve Cashflow in a Plywood and Interior Materials Business

The Final Report for the BDM Capstone Project

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1. Executive Summary

The business, a regional distributor of home improvement goods, relies on a hybrid sales model involving contractors and direct End Customers (ECs). While contractors drive indirect EC sales, this channel is fragile, heavily dependent on ~20 top individuals. The business also faces recurring liquidity strain from unregulated credit sales and inventory bloating due to poor demand alignment.

Data from FY25 (April '24–March '25) was collected, covering sales, procurement, and contractor behavior. Contractor performance data, which the business maintains by organizing a 600+ informal contractors network into A, B, and C lists on the basis of individual sales target hit. That is, how much sales did each contractor enable, along with contractor payouts.

Descriptive statistics, correlation analysis, regression models, channel-mix simulations, EOQ modeling, and what-if analysis were employed. Bimodality in the EC segment hinted possibility of subsegments within the channel. Variability in AOVs of organic vs contractor-influenced EC sales from the 18 Estimate Memo samples further augmented the belief.

Payment flows were traced using Sankey diagrams, while contractor fragility was quantified using churn risk metrics and inventory impact analysis. Contractors triggered ₹30,000+ EC orders while placing only ₹1,000 direct orders—making them high-leverage enablers, not direct value creators. EOQ modeling revealed a starkly high over-ordering rate, and 35% of UPI sales (~₹17L) were credit-based, explaining cash flow stress. Removing the top 20 contractors risks a ~50% drop in contractor-led revenue.

Recommendations include formalizing contractor SLAs, monthly data review of contractors' performance instead of yearly (Diwali bonuses rollout). Digitizing End Customer acquisition, enforcing credit limits, and adopting seasonal EOQ logic. If implemented, these actions could reduce credit exposure by ₹6–11L, save ₹2.6L in ordering costs, and improve inventory turnover. This positions the business for stable, scalable operations by FY26.

2. Detailed Explanation of Analysis Process/Method

Data Collection, Extraction, Cleaning, Feature Engineering & Domain-Guided Adjustments

Sales, purchase, and contractor data were digitized using Windows 11’s “Scan Text” and a custom Python script, with manual entry for smaller invoice sets. Data cleaning included rounding, OCR correction, deduplication, and column standardization. Feature engineering added fields like Buyer Role, Type, Payment Method, and Description. For deeper segmentation, 9 samples each of “organic” and “contractor-influenced” end-customer estimate memos were manually extracted with memo IDs, dates, and amounts.  Estimate Memos

Operational insights were gathered from the business owner to address missing fields. Key disclosures included:

Table A. Business Owner Disclosures and Working Assumptions

#	Disclosure	Details
1	End Customer Influence Attribution	~80% End Customer Sales are Contractor-Influenced
2	Credit Sales in Unregistered Segment	Approximately 30–40% of unregistered sales are made on credit
3	Monthly Inventory Wastage	Total: 15–20%; Deadstock: 13–15%, Damaged: ~5% (Laminates Only)
4	Maximum Credit Delay Observed	Up to 2 years in extreme cases (Undocumented)
5	Outstanding Credit Amount	₹16–18 lakh currently stuck in market receivables
6	Nature of Registered Sales	All are B2B with full upfront payment (no credit exposure)
7	Operational Time Spent on Collections	An estimated 10% of daily operations go into credit recovery
8	Estimated Revenue Loss due to Credit	Approximately 5% loss attributed to delayed or defaulted credit
9	Ordering Cost	Far-off locations: 2.5-3%; Near-off Locations: 1.5-2% of Purchase Amt.

2.1 Derivation of Core Inventory Data Variables

Procedure and Formulas Used For Deriving Key Inventory Data		Variable Definitions
Step 0: Weighted Averages $\bar{P}_s = \frac{\sum S}{\sum Q_s}, \quad \bar{P}_p = \frac{\sum P}{\sum Q_p}$	Step 5: Stock Turnover Ratio $STR_t = \frac{Q_s}{\frac{O_t + C_t}{2}} \times 100$	$\bar{P}_s$ – Weighted Average Selling Price (₹/unit) $\bar{P}_p$ – Weighted Average Purchase Price (₹/unit) S – Total Sales Amount (₹) P – Total Purchase Amount (₹) Q_s – Sales Units for the month Q_p – Purchase Units for the month S_m – Monthly Sales Amount (₹) P_m – Monthly Purchase Amount (₹)
Step 1: Monthly Sales & Purchase Units $Q_s = \frac{S_m}{\bar{P}_s}, \quad Q_p = \frac{P_m}{\bar{P}_p}$	Step 6: Wastage% $W\% = \frac{W}{O_t + Q_p} \times 100$	

Step 2: Wastage Unit $W = w \times (Q_s + Q_p)$ (Assumed $w=18\%$)	Step 7: Holding Cost $H = r \times \frac{O_t + C_t}{2} \times \bar{P}_p$ (Where $r = \frac{20\%}{12} = 0.0167$, $\bar{P}_p \approx ₹247$)	w – Wastage Rate (assumed 18%) W – Wastage Units O_t – Opening Stock in month t O_{Apr} – Opening Stock in April (initial stock assumption) C_t – Closing Stock in month t STR_t – Stock Turnover Ratio in month t H – Holding Cost r – Monthly Holding Rate (assumed 1.67%) $\bar{P}_p$ – Avg. Selling Price (₹400 assumed) $W\%$ – Monthly Wastage Percentage $MA_3(Q_s)$ – 3-Month Moving Average of Sales Units $Excess_t$ – Excess stock (relative to demand trend)
Step 3: Opening Stock $O_{Apr} = 1.2 \times \text{Avg}(Q_s)$ $O_t = C_{t-1}$	Step 8: Excess Stock $Excess_t = \frac{C_t}{MA_3(Q_s)}$ (3-month Moving Average of Sales Units)	
Step 4: Closing Stock $C_t = O_t + Q_p - Q_s - W$		

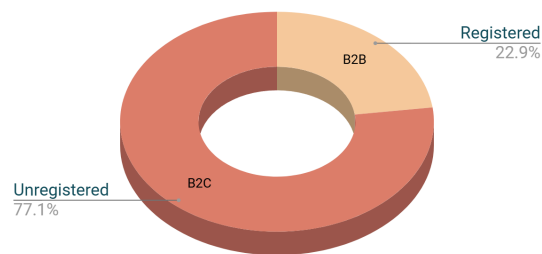
Table 2.1: Derivation of Inventory Data Variables (To view a simpler version, refer to the **Data Derivation Sheet** inside the *Cleaned & Estimated* page of the [Inventory Data \(Apr-24 - Mar-25\) \(Estimated\)](#))

2.2 Essential Diagnostic Approaches

Descriptive statistics were computed for sales across buyer roles—B2B (Registered), Contractors, and End Customers—to understand transactional behavior and revenue distribution. Summary metrics, including count, mean, median, standard deviation, and range, are presented in the adjacent table.

Segment	Count	Min (₹)	Median (₹)	Mean (₹)	Max (₹)	Std. Dev (₹)	Total (₹)
B2B (Registered)	105	240	30,200	29,160	2,44,213	26,733	30.6L
Contractor	1336	19	1,462	1,000	5,000	1,320	19.5L
End Customer	500	5,010	16,680	12,500	49,982	11,381	83.4L

Table 2.2: Order Value Distribution Summary by Sales Channel



B2B transactions have the highest average order value (₹29,160) but also the greatest variance (std. dev ₹26,733), with a long right-skewed tail, typical of project-based high-ticket purchases. Contractors show a bimodal distribution, suggesting two distinct behavioral clusters: micro-buyers and mid-scale operators. Despite their lower average (₹1,000), contractors drive high volume. End Customers exhibit a narrower, right-skewed distribution centered around ₹15,000–₹20,000, indicating relatively consistent purchase behavior.

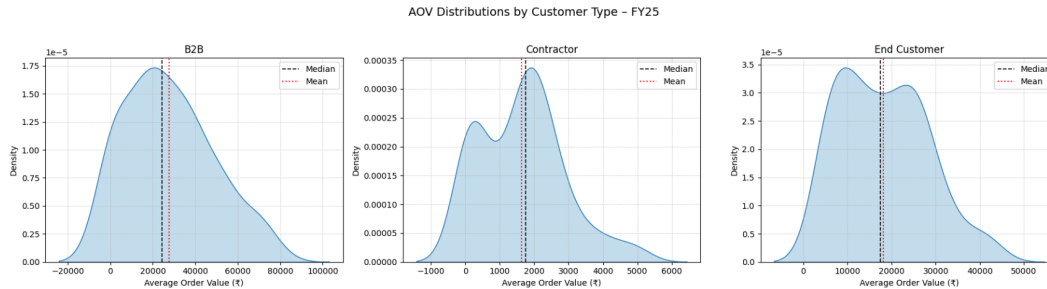


Figure 2.2 [a], [b], [c]: Kernel Density Estimates (KDE Plot) for AOV Distributions by Buyer Segments

Interestingly, while B2B orders are high-value, they contribute only ~23% of total revenue. Over 75% is driven by unregistered sales, largely from End Customers and Contractors transacting via UPI. This underscores that overall revenue is powered by high-frequency, low-ticket purchases.

These insights on central tendency, variance, and distribution shapes provide a quantitative foundation to assess credit risk, inventory turnover, and buyer segmentation strategies discussed in later sections. [🔗 Descriptive Statistics for principal segments.ipynb](#)

2.2.1 Diagnosing Channel Profitability and Structural Risk in the Contractor-Influenced End-Customer Sales

2.2.1.1 Evidence of Behavioral Heterogeneity Within End Customers

In the initial phase of analysis, a one-way ANOVA was conducted across the three principal sales channels—B2B, Contractor, and End Customer—to evaluate their relative financial contributions. This test revealed that the End Customer segment demonstrated the highest average order value (AOV), prompting further inquiry into whether this performance was uniformly distributed or skewed by latent subgroups within. However, Table A: point [1] warranted deeper scrutiny. Hence, a micro-sample of 18 estimate memos (9 organic and 9 contractor-driven end customer sales [🔗 Organic vs Contractor-Influenced](#)) was collected and analyzed to explore potential behavioral differences.

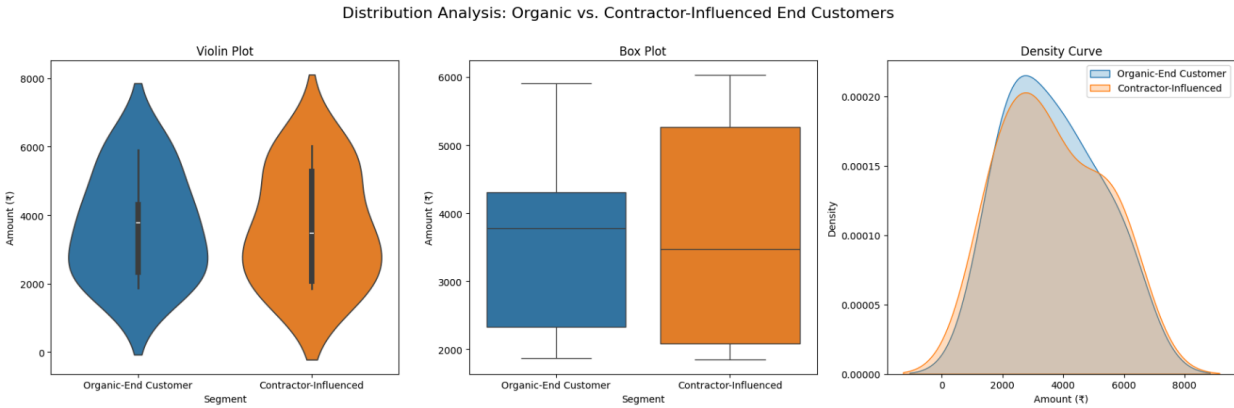


Figure 2.2.2.1: Distribution Analysis: Organic vs. Contractor-Influenced End Customers

This illustration is based on a limited sample (n=9 each) and is intended only for exploratory comparison, not inferential generalization.

Despite the small size, contractor-driven sales showed higher average order values and wider dispersion. Though not statistically generalizable, this poses the **existence** of two distinct buyer behaviors within the end customer group, which is also observed in the bimodality of Figure 2.2 [c]. These trends were used to parameterize a Monte Carlo simulation for further exploration.

[🔗 Descriptive Statistics for Subsegmented EC.ipynb](#)

2.2.1.2 Probabilistic Subsegmentation and Statistical Validation of End-Customer Subtypes

To validate whether these observed behavioral distinctions scaled across the full dataset, a **Monte Carlo simulation** was employed. Using Table A: point [1], the dataset was repeatedly subsegmented into **Contractor-Influenced ECs** and **Organic ECs** across 1,000 randomized iterations. From these simulations, a representative sample was selected to serve as the basis for statistical testing and visualization. Descriptive statistics, Levene's test for equality of variances, and independent t-tests were used to compare the two EC subgroups.

[🔗 Statistical Tests on EC Subsegments.ipynb](#)

2.2.1.3 ANOVA Test Re-Run for Profitability Analysis in New Defined Customer Segments

Finally, a revised ANOVA was run across four discrete segments: B2B, Contractor, Contractor-Influenced End Customers, and Organic End Customers. Given the results of Levene's test indicating unequal variances among the groups, Welch's ANOVA was employed as a robust alternative to the classical ANOVA. To further examine pairwise differences while

accounting for variance heterogeneity and unequal sample sizes, the Games-Howell post-hoc test was subsequently applied. [🔗 Statistical Tests on EC Subsegments.ipynb](#)

2.2.1.4 Channel Mix Simulation and Attrition Risk Model

Next, simulated 1000 "days" or "sale cycles" with, say, 100 ECs per cycle. For each cycle, ECs were divided per scenario % between Contractor-Influenced and Organic.

EC_Type	count	mean	std	min	25%	\
Contractor-Influenced	400.0	16848.9725	11602.785125	5010.0	7800.0	
Organic	100.0	16003.7100	10474.640923	5100.0	7195.0	
	50%	75%	max			
EC_Type						
Contractor-Influenced	12500.0	22550.0	49982.0			
Organic	12000.0	20000.0	41100.0			

Scenario	Contractor-Influenced EC %	Organic EC %
A	80%	20%
B	50%	50%
C	30%	70%
D	0%	100%

Figure 2.2.1.4: EC Segment AOV Characteristics and Channel Mix Scenarios for Revenue Simulation

For each EC, randomly sampled a revenue from a normal distribution using the mean and standard dev of the respective subsegment. [🔗 Channel Mix Simulation.ipynb](#)

Addressing the three core dimensions of risk: revenue attrition, commission overhead, and recovery lag, **Contractor Dependency Risk Modeling** was conducted. Quantified potential revenue loss under varying levels of referral drop-off, estimated the commission overhead associated with contractor-mediated sales, and projected the time required to recover lost volume through organic growth alone. Inputs included observed average order values, referral share estimates, payout ranges, and current organic acquisition rates. [🔗 Attrition Risk Model.ipynb](#)

2.2.2 Diagnosing Contractor Network Fragility and Revenue Concentration Risk within Top Performing Contractors (A-List)

2.2.2.1 Structural Concentration and Contribution Inequality in the Contractor Channel (CDI + Lorenz/Gini)

To assess concentration risk within the contractor network, two complementary analyses were conducted on the A-list segment (top 67 contractors, representing ~11% of the base). First, the **Contractor Dependency Index (CDI)** was calculated using the Herfindahl–Hirschman Index

(HHI) methodology. Each contractor's revenue share was squared and summed to compute the HHI, which was then subtracted from 1 to yield the CDI. A lower CDI indicates that a few contractors dominate revenue, flagging fragility in the network's structure.

To complement this numerical measure, a **contribution analysis** was performed by calculating the cumulative revenue share of the top 5, 10, 20, and 50 contractors. These data points were used to plot a **Lorenz curve** and derive a **Partial Gini coefficient**, quantifying inequality within this high-performing subset. While CDI provided a scalar view of concentration, the Lorenz-Gini approach visualized distributional skew, together offered a comprehensive view of contractor dependency risks.

2.2.2.2 Stress Testing Contractor Channel Fragility Using Probabilistic and Sensitivity-Based Revenue Modeling

To quantify the potential downside of overreliance on a small subset of top-performing contractors, a stress testing framework was implemented on the A-list pool ($n = 67$), which collectively generated ₹1.57Cr in revenue (Nov'23-Oct'24). Though this group represents only ~11% of the total contractor base (~600+), their contribution warranted focused risk modeling. The simulation was conducted in two tiers:

2.2.2.2 [i]: Binary Churn Scenarios

Revenue loss was estimated by hypothetically removing the top 5, 10, and 20 contractors based on their contribution. The corresponding drop in total A-list revenue was calculated for each scenario to reflect abrupt attrition events. To quantify the uncertainty in these estimates, 95% confidence intervals were constructed using the standard error of the proportion. Additionally, a one-sample t-test was run against a null hypothesis of 0% loss to evaluate the statistical significance of observed dependency.

2.2.2.2 [ii]: Partial Attrition Scenarios (Sensitivity Analysis)

Recognizing that contractor disengagement may not be absolute, a sensitivity analysis was layered in. For each top-X contractor group (5, 10, and 20), revenue contributions were proportionally reduced by 25%, 50%, and 75% to simulate partial drop-offs. This probabilistic

extension enabled modeling of revenue exposure under a spectrum of disengagement severities, offering a more nuanced view of downside risk in realistic conditions.

2.2.2.3 Impact of Contractor Reliance on Channels and Inventory

To explore how overreliance on contractors influences other channels and inventory, correlations were computed using FY25 monthly Inventory data (Apr-24 to Mar-25).

2.2.3 Diagnosing Seasonal Demand Volatility by Designing Segment-Level Demand Stability Scorecard

2.2.3.1 Segment-wise Volatility Assessment via CV and MAD

The initial ANOVA test confirmed that End Customers were the most profitable, but also the most inconsistent, with B2B showing similar volatility. Contractors, in contrast, exhibited steadier sales. However, ANOVA only showed differences in average order values, not how erratic demand was over time. To fill this gap, volatility was assessed using two metrics: **Coefficient of Variation (CV)** for relative fluctuations and **Mean Absolute Deviation (MAD)** for absolute ones, each offering a different perspective on demand stability.

2.2.3.1 [i]: Coefficient of Variation (CV): Volatility Relative to Scale

The CV, calculated as the ratio of the standard deviation to the mean (multiplied by 100), quantifies relative volatility, allowing comparison between segments of different scales. Monthly sales data was aggregated for each segment (B2B, Contractor, End Customer), and the population standard deviation (STDEV.P) was used, treating FY25 as a closed dataset.

2.2.3.1 [ii]: Mean Absolute Deviation (MAD): Volatility in Actual Terms

Next, to complement CV's relative measure, MAD was used to capture absolute monthly demand deviations from the mean. By averaging these absolute differences, MAD offers a robust, outlier-resistant view of typical fluctuation magnitude—expressed in original sales units—revealing how much demand strays from the norm, regardless of direction.

2.2.3.2 Time Series Pattern Detection: Growth Trends & Autocorrelation

While CV and MAD measure the magnitude of demand fluctuations, they don't reveal direction or rhythm. To capture these temporal dynamics, two time-series tools were used: **Sales Growth Rate** (for month-over-month trends) and **Autocorrelation Function (ACF)** (for recurring patterns or cyclicalities).

2.2.3.2 [i]: Segment-wise Sales Growth Rate

For each month, the sales growth rate was computed as the percentage change in sales compared to the previous month. Formula used:

$$\text{Growth Rate}_t = \left(\frac{\text{Sales}_t - \text{Sales}_{t-1}}{\text{Sales}_{t-1}} \right) \times 100$$

These calculations were performed using the monthly sales data from FY25, with the first month (April) excluded due to the lack of a prior comparison point.

2.2.3.2[ii]: Autocorrelation Function Analysis

Autocorrelation coefficients for B2B, Contractor, and End Customer segments were computed by correlating monthly demand with its lagged values (1–6 months), using the CORREL function and adjusting for reduced sample size at higher lags.

Lag	B2B ACF	Contractor ACF	End Customer ACF
1	-0.0605	0.2855	0.0768
2	-0.7494	-0.0790	-0.1875
3	0.0132	-0.0479	0.1718
4	0.4336	-0.1765	0.0206
5	-0.5259	0.1029	-0.1146
6	-0.0618	0.1154	0.1410

Table 2.2.3.2 [e]: Autocorrelation Coefficients (ACF) by Segment Across Lags (Lag 1–6, FY25)

The coefficients (ranging from -1 to 1) indicate the strength and direction of demand patterns—positive for repetition, negative for reversal. Though standard practice limits analysis to 3-month lags ($n/4$), it was extended to 6 months ($n/2$) to detect potential semi-annual cycles relevant for inventory planning.

2.2.3.3: Segment-Level Demand Stability Scorecard

Segment	Std Dev	CV (%)	MAD	Sales Volatility	ACF Lag 1
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B2B	107359	42	92883	87.54%	-0.0605
Contractor	36644	23	32342	36.16%	0.2855
End Customer	260891	38	276187	68.09%	0.0768

Table 2.2.3.3 [a]: Raw Demand Volatility Metrics Across Customer Segments

Each metric, except ACF, was scaled using min-max scaling so that higher volatility $\rightarrow$ lower stability. For ACF, since a higher absolute value means more stability:

$$\text{ACF Norm} = \left(\frac{|\text{Current}| - \text{Min}}{\text{Max} - \text{Min}} \right) \times 100$$

Metric	Min	Max	Segment	Norm.S td	Norm. CV	Norm.M AD	Norm.V ol	Norm.A CF
Std Dev	36,644	260,891	B2B	0.679	0	0.789	0	0
CV (%)	23	42	Contractor	1	1	1	1	1
MAD	32,342	276,187	End Cust.	0	0.2105	0	0.377	0.0725
Sales Volatility %	36.16	87.54						
abs(ACF Lag 1)	0.0605	0.2855						

Table 2.2.3.3 [b]& [c]: Min–Max Scaling Bounds for Demand Stability Metrics and Normalized Segment-Wise Stability Scores Across Volatility Metrics

To quantify and compare demand stability across customer segments using a composite index built from multiple volatility metrics, two custom indices were developed to address separate operational risks:

Stability Index A: Prioritizes inventory management accuracy.

$$\text{Stability Index}_A = 100 \times (0.3 \cdot \text{CV}_{\text{norm}} + 0.25 \cdot \text{MAD}_{\text{norm}} + 0.1 \cdot \text{STD}_{\text{norm}} + 0.25 \cdot \text{Vol}_{\text{norm}} + 0.1 \cdot \text{ACF}_{\text{norm}})$$

Stability Index B: Focuses on channel predictability.

$$\text{Stability Index}_B = 100 \times (0.25 \cdot \text{CV}_{\text{norm}} + 0.10 \cdot \text{MAD}_{\text{norm}} + 0.10 \cdot \text{STD}_{\text{norm}} + 0.25 \cdot \text{Vol}_{\text{norm}} + 0.3 \cdot \text{ACF}_{\text{norm}})$$

Segment	Stability Index A (Inventory Focus)	Stability Index B (Channel Predictability)
B2B	26.52	14.68
Contractor	100.00	100.00

End Customer	16.07	16.86
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Table 2.4.2.3 [d]: Composite Demand Stability Indices by Segment (Inventory vs. Channel Perspective)

2.2.4 Responsive Inventory Optimization and Seasonality-Aware EOQ Modeling

2.2.4.1 Inventory Efficiency Analysis: Turnover Ratios, Excess Stock, and Holding Costs

To assess SK Timber's inventory alignment with demand cycles, key efficiency metrics, including monthly turnover ratio (Table 2.2: Step 5), wastage (Table 2.2: Step 6), holding cost (Table 2.2: Step 7), and excess stock percentage (Table 2.2: Step 8), were analyzed.

2.2.4.2 Mapped Volatility to Inventory KPIs

Now, to align demand volatility with inventory inefficiencies, it was required to answer whether sales months are followed by high stock buildup, whether high-demand months are preceded by inadequate purchasing, and whether high holding costs correlate with low turnover and low sales.

2.2.4.3 Seasonal EOQ Modelling, System Benchmarking, and Annual Cost Simulation

To optimize procurement efficiency and minimize inventory-related costs, the **Economic Order Quantity (EOQ)** model was applied using sourcing-specific operational assumptions tailored to SK Timber's context. EOQ helps identify the optimal order size that minimizes the **combined costs of ordering and holding inventory**, balancing procurement frequency with storage capacity.

The EOQ was calculated using the classical formula:

$$EOQ = \sqrt{(2DS/H)}$$

Where:

- **D** = Annual purchase demand (sum of monthly purchase units)
- **S** = Ordering cost per order (as a percentage of average purchase value per order)
- **H** = Annual holding cost per unit, estimated at 20% of the average unit purchase price

Recognizing sourcing cost differences, EOQs were calculated separately: **1.75% order cost for near-off** (within Aligarh) and **2.75% for far-off** sources. Holding cost (H) was held constant due

to centralized storage. To account for **seasonal demand**, EOQs were further segmented into **high, mid, and low demand months**, enabling responsive procurement strategies.

The optimized EOQs were then benchmarked against the current system, revealing inefficiencies in order frequency and size. A simulation of **annual ordering costs** under both approaches demonstrated the cost-saving potential of switching to an EOQ-based model.

2.2.5 Lightweight Credit Risk Assessment via Payment Behavior and Cashflow Stress Indicators

Data were highly limited within the credit context. Hence, to uncover hidden credit risk affecting cash flow, a lightweight proxy analysis was conducted using payment methods as indicators of credit exposure. The working hypothesis treated **CASH payments** as immediate settlements (low credit risk), while **UPI payments** potentially included deferred or informal credit arrangements, especially for contractors and end customers. This assumption was later confirmed through qualitative input from the business owner.

Payment Method	Buyer Role	Count	Avg. Amt.(₹)	Median (₹)	Max (₹)	Total (₹)	Share (%)
Cash	Contractor	295	₹678	₹400	₹4,366	₹2,00,084	1.50%
	End Customer	1	₹14,500	₹14,500	₹14,500	₹14,500	0.11%
UPI	B2B	105	₹29,160	₹30,200	₹2,44,213	₹30,61,811	22.93%
	Contractor	1041	₹1,684	₹1,300	₹5,000	₹17,52,548	13.12%
	End Customer	499	₹16,684	₹12,500	₹49,982	₹83,25,460	62.34%

Table 2.2.5 Sales Breakdown by Payment Method and Buyer Role

Sales transactions were broken down by **buyer role** and **payment type**, capturing metrics such as count, average amount, median, and total sales as shown in the table below. The goal was to quantify how much revenue was tied to **potentially delayed payments** and whether any systematic financial risk could be inferred.

2.2.6 Risk Prioritization Using Failure Mode and Effects Analysis (FMEA)

To prioritize operational risks, a Failure Mode and Effects Analysis (FMEA) was conducted. Each failure mode was assessed across three dimensions:

- **Severity (S)** – How serious the impact is (1–10)
- **Occurrence (O)** – Likelihood of it happening (1–10)
- **Detection (D)** – Likelihood of detecting it before it causes harm (1–10)

The **Risk Priority Number (RPN)** is calculated as:

$$RPN = S \times O \times N$$

Higher RPNs indicate more critical risks needing urgent action.

Failure Mode	S	O	D	RPN	Notes / Impact
Loss of top 20 contractors	9	7	5	315	~49% of contractor-driven revenue at risk; long recovery time (4+ months)
Rising credit defaults from ECs	8	6	4	192	₹16–18L in receivables stuck; affects cash flow and liquidity
Seasonal overstocking	7	5	4	140	Excess stock peaked at 148%; high holding costs, low turnover
Inventory misalignment due to demand lag	6	6	3	108	Procurement in March ignored low sales, triggering waste
Contractor disengagement (non-A-list)	5	5	5	125	Reduced redundancy in the contractor channel undermines future resilience
Delayed detection of churn risk	6	4	6	144	Annual reviews are too slow; churn could've been prevented via monthly tracking

Table 2.2.6: FMEA Summary Table

The **highest risk (RPN 315)** is over-reliance on top contractors, highlighting the need for retention mechanisms and diversification. Credit default risk (RPN 192) also ranks high, warranting formal credit controls and digital ledger integration. Overstocking and procurement inefficiencies (RPNs 140 & 108) point to the need for real-time, season-aware inventory management.

3. Results and Findings

3.1 Summary of Key Findings from Segment Decomposition and Channel Dependency Analysis

While preliminary analysis identified End Customers as the most profitable group, subsequent segmentation revealed that **contractor-mediated sales disproportionately drove this uplift**. Upon decomposing the EC segment using a heuristic Monte Carlo framework, revised ANOVA and Games-Howell post hoc test confirmed that B2B is the true leader in average order value.

3.1.1 Levene's Test, t-test, and Descriptive Statistics

The test yielded a p-value of **0.552**, failing to reject the null hypothesis and indicating that the assumption of homogeneity of variances within EC subsegments (contractor-influenced vs organic) **holds**. Consequently, a **standard independent t-test ($p = 0.481$)** was deemed appropriate. The result confirmed that the difference in average order value between the two subsegments was **not statistically significant**. [🔗 Statistical Tests on EC Subsegments.ipynb](#)

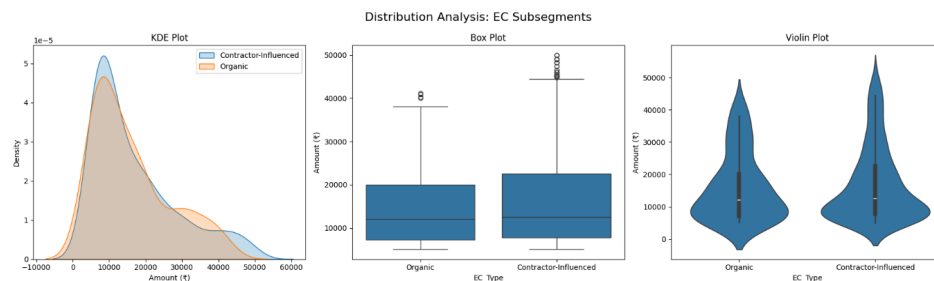


Figure 3.1.1[a], [b], & [c]: Kernel Density Plot, Box Plot, and Violin Plot for EC Subsegmentation Distribution Analysis

KDE: Contractor-Influenced curve is slightly *right-shifted* (higher AOV). Both curves heavily overlap, especially in the ₹5K–₹25K range, aligning with the t-test result ($p = 0.481$): not statistically different, but practically meaningful. A visible bimodal tail in Contractor-Influenced ECs indicates some large-value orders skewing the distribution, suggesting higher upside potential in that subsegment.

Box Plot and Violin Plot: Contractor-Influenced ECs exhibit a higher median and a wider spread compared to Organic ECs, with a concentration of **upper outliers** in the ₹40K–₹50K range, likely representing bulk or project-driven orders. The fatter upper tail in the violin plot

confirms a right-skewed distribution, indicating more unpredictability but greater upside potential. In contrast, Organic ECs are more centered and symmetric, suggesting lower but more consistent order values.

This highlights a key trade-off: Contractor-Influenced = higher variability and profit potential, while Organic = reliability with a capped ceiling.

When extending the analysis to all four principal segments — **B2B, Contractor, Contractor-Influenced EC, and Organic EC** — Levene’s test returned an extremely low p-value (2.42×10^{-159}), clearly rejecting the null hypothesis of equal variances. Given this violation of the homogeneity assumption, a **Welch’s ANOVA** was employed, followed by a **Games-Howell post-hoc test** to explore pairwise differences robustly.

Source	ddof1	ddof2	F	p-unc	np2
0 Segment	3	225.470235	332.639202	1.717917e-82	0.496791

Figure 3.1.1[d]: Welch’s ANOVA on All Four Segments (B2B, Contractor, Contractor-EC, Organic EC)

With $F(3, 225.47) = 332.64$, $p < 0.00001$, high statistical significance is observed in the Welch’s ANOVA, and with the large effect size (partial $\eta^2 \approx 0.5$)—nearly half the variance in order values is explained by the segment type.

Segment A	Segment B	Mean AOV A (₹)	Mean AOV B (₹)	Difference (₹)	p-value	Effect Size (Hedges’ g)
B2B	Contractor	₹29,160	₹1,462	₹27,699	< 0.001	3.79
B2B	Contractor-Influenced	₹29,160	₹16,849	₹12,311	< 0.001	0.77
B2B	Organic	₹29,160	₹16,004	₹13,156	< 0.001	0.64
Contractor	Contractor-Influenced	₹1,462	₹16,849	₹15,387	< 0.001	-2.71
Contractor	Organic	₹1,462	₹16,004	₹14,542	< 0.001	-4.79

Table 3.1.1: Pairwise Segment Comparison: Mean Order Value Differences and Effect Sizes (Games-Howell Post-Hoc Results)

Effect sizes indicate B2B outperforms all others with a margin bordering on decisive (Hedges’ $g > 3.7$ in some comparisons). Contractor-Influenced ECs have higher AOVs than Organic ECs consistently, but not statistically significantly. Although contractor-influenced EC sales are not statistically more profitable than organic sales, they do have higher AOVs than organic. This

implies these customers likely come in with larger projects, higher budgets, or greater confidence because they were referred.

3.1.2 Channel Mix Simulation: Summary & Implications

Scenario	Contractor-Influenced %	Organic %	Mean Revenue (₹)	Std Dev (₹)	5th–95th Percentile Range
A (80-20)	80%	20%	₹16.64L	₹1.13L	₹14.79L – ₹18.53L
B (50-50)	50%	50%	₹16.40L	₹1.11L	₹14.49L – ₹18.14L
C (30-70)	30%	70%	₹16.22L	₹1.06L	₹14.40L – ₹17.90L
D (0-100)	0%	100%	₹15.97L	₹1.00L	₹14.33L – ₹17.60L

Table 3.1.2: Simulated Revenue Outcomes Under Varying Contractor Influence Scenarios

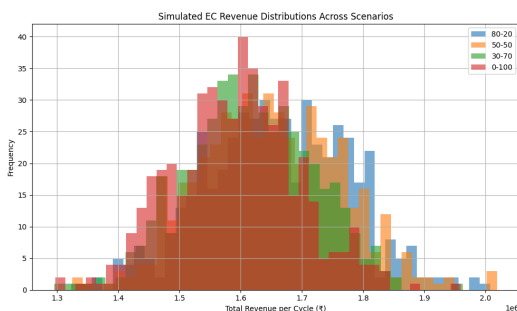


Figure 3.1.2: Simulated EC Revenue Distributions Across Scenarios

Revenue Drops with Contractor Attrition:

Moving from 80% to 0% contractor-influenced ECs causes a drop of nearly ₹67K per sales cycle (₹16.64L → ₹15.97L). That's ~4% loss in top-line EC revenue per 100 transactions. **Volatility**

Narrows with Organic Dominance: Std deviation

drops from ₹1.13L (80-20) to ₹1.00L (0-100),

which makes sense — the contractor channel injects *both* upside and unpredictability. Risk Envelope Shrinks: The 95th percentile revenue in the 80-20 mix hits ₹18.5L, while in a fully organic world, it maxes out around ₹17.6L. Business would be capping its upside if it abandons the contractor channel. [Channel Mix Simulation.ipynb](#)

3.1.3 Findings from Attrition Risk Model (Revenue Loss Estimation)

Reveals critical risks associated with the current sales channel structure. If all contractor referrals were to cease, the business would face a revenue loss of approximately ₹16.8 lakh per sales cycle. Even a moderate attrition of 30% would result in a revenue shortfall exceeding ₹5 lakh, an unsustainable hit for small to medium enterprises.

Attrition Rate	Estimated Revenue Loss (₹)
0	0
1	168490
2	336980
3	505470
4	673960
5	842450
6	1010940
7	1179430
8	1347920
9	1516410
10	1684900

At 9% commission, contractor payout = ₹121,500
At 10% commission, contractor payout = ₹135,000
At 11% commission, contractor payout = ₹148,500
At 12% commission, contractor payout = ₹162,000
At 13% commission, contractor payout = ₹175,500
At 14% commission, contractor payout = ₹189,000

Time to recover: 4.0 months
Revenue loss until recovery: ₹1,280,320

Figure 3.1.3 [a], [b], & [c]: Revenue Attrition, Commission Overhead, and Recovery Timeline

Additionally, the contractor channel imposes a recurring commission overhead, with payouts ranging from ₹1.2 lakh to ₹1.9 lakh per cycle, effectively consuming 9–14% of the top-line revenue. In the event of complete contractor disengagement, rebuilding the lost transaction volume through organic

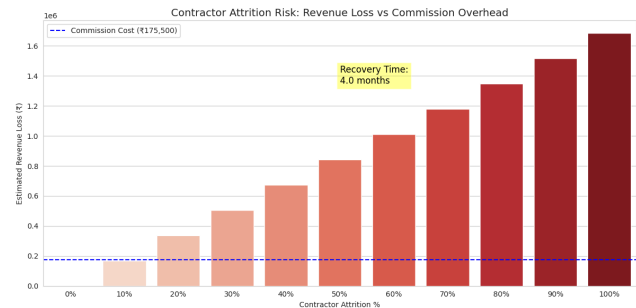


Figure 3.1.3 [d]: Bar Chart visualizing Contractor Attrition Risk: Revenue Loss vs Commission Overhead

end-customer acquisition would require an estimated four months. During this recovery lag, the business stands to forgo an additional ₹12.8 lakh in potential revenue, underscoring the strategic vulnerability of overreliance on contractor-mediated sales. [Attrition Risk Model.ipynb](#)

3.2 Assessing the Contractor Network Fragility and Contractor-influenced Revenue Concentration Risk

3.2.1 Structural Concentration in the Contractor Channel

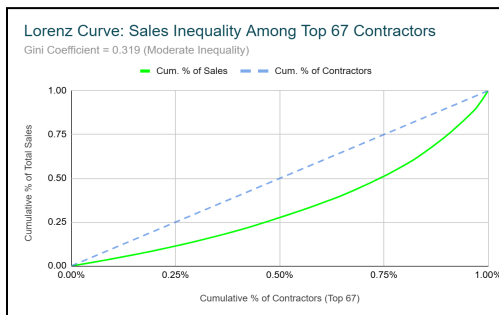


Figure 3.2.1 [a]: Visualizing Sales Inequality in A-List using Lorenz Curve (Gini=0.329)

Despite having a seemingly high CDI (0.978), the top 5–20 contractors are disproportionately contributing to revenue, exposing a structural fragility beneath the surface of a “diverse” A-list. This was evident in the Lorenz Curve.

([Contractor Performance Data](#)).

3.2.1 [i] Findings from Binary Churn Simulation

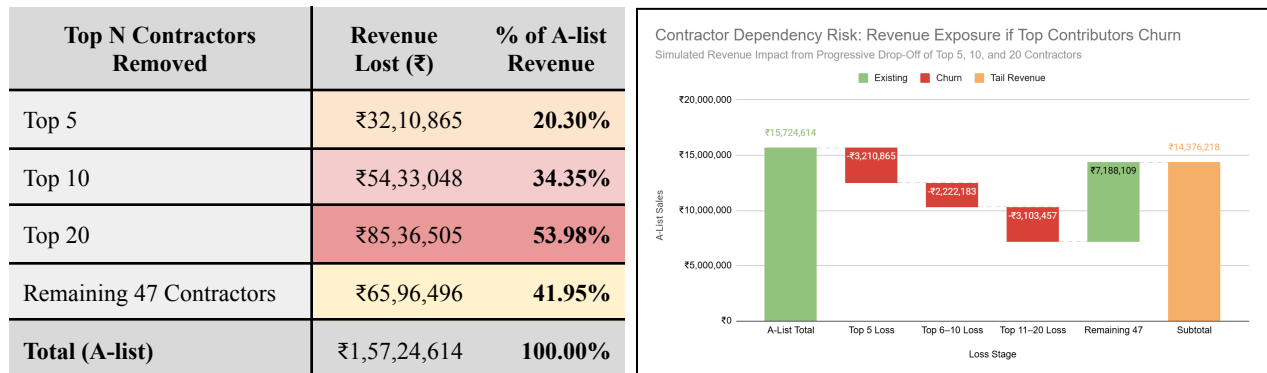


Figure 3.2.1 [b] & [c]: Visualizing Sales Inequality in A-List using Lorenz Curve (Gini=0.329)

This chart highlights that if our top 10 contractors were to churn, we'd lose **nearly 34%** of contractor revenue. Removing the top 20 leaves us with just **~46%** of the A-list's value, meaning over half of the contractor channel depends on a very thin slice of contributors.

Significance Tests on Churn Analysis:

Top N Contractors	% Revenue Lost	95% CI (%)	Revenue Lost (₹)	95% CI (₹)
Top 5	20.30%	10.67-29.93	₹3,192,077	₹1,677,927 - ₹4,706,227
Top 10	34.35%	23.98-44.72	₹5,401,896	₹3,770,176 - ₹7,033,616
Top 20	53.98%	42.04-65.92	₹8,489,231	₹7,117,672 - ₹10,360,790

Table 3.2.1: Confidence Intervals for Revenue Loss and Significance Test

A one-sample t-test confirms significant revenue loss ($p < 0.001$) when top contractors churn, with t-values of 4.13 (top 5), 5.92 (top 10), and 8.86 (top 20), rejecting the null hypothesis of no impact. This underscores the channel's fragility, as losing key contributors could disrupt over half the A-list value. However, these estimates are based on A-list data (Oct-Nov 2024) as a proxy, with limitations due to single-period sampling. [📊 Contractor Performance Data](#)

3.2.1 [ii] Scenario Results of Sensitivity-based Contractor Churn Simulation

Simulated revenue loss scenarios reveal stark vulnerabilities in the contractor-driven sales channel. Under the A-list revenue baseline (₹1.57 crore), a partial drop of just 50% in sales enabled by the top 10 contractors would wipe out over ₹27 lakh, amounting to 17.28% of A-list

revenue. Escalating the disengagement to the top 20 contractors at a 75% drop results in a revenue loss of ₹64 lakh, eliminating over 40% of this elite tier’s contribution.

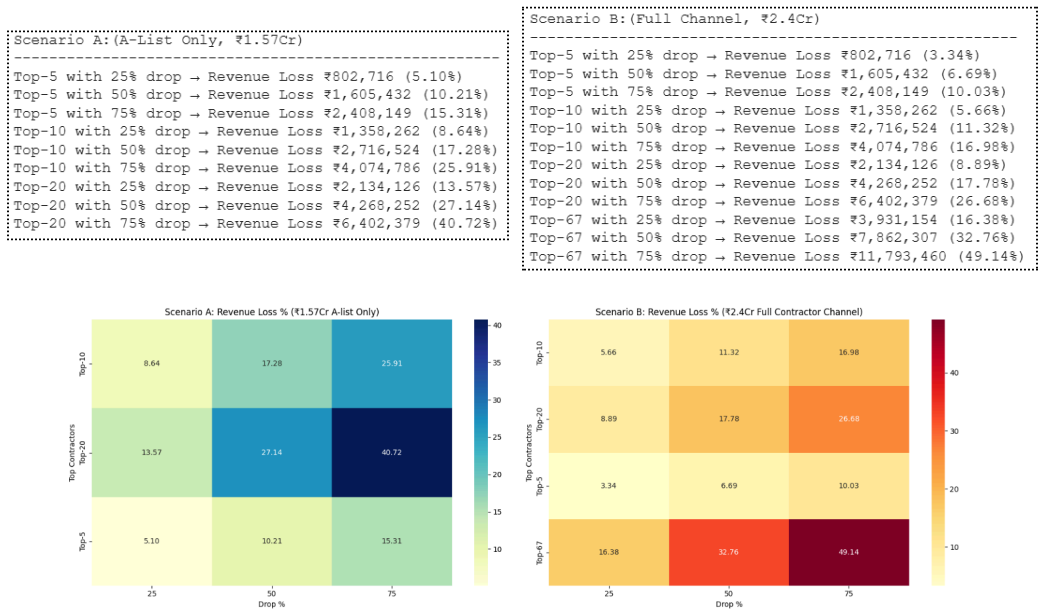


Figure 3.2.1 [d], [e], & [f]: Sensitivity Analysis of Contractor Channel: Simulated Revenue Impact Under Partial Attrition Scenarios (A-List and Combined estimate of A, B, and C list)

When the broader contractor base (₹2.4 crore **estimated total revenue**) is considered, the damage scales but appears slightly more contained percentage-wise. A 75% disengagement from the top 20 contractors still erodes ₹64 lakh, but now translates to 25.61% of total contractor revenue. Strikingly, a 75% disengagement of the entire A-list (top 67 contractors) results in a ₹1.18 crore hit, nearly half of the total contractor-enabled revenue (47.17%). These results highlight a critical pattern: despite the appearance of diversification (67 contractors in the A-list), the sales engine is disproportionately powered by a thin layer at the top. Even partial disengagement among top contributors could cripple performance, indicating that contractor reliance isn't just a fragility—it's a strategic risk. [🔗 Binary and Gradual Contractor Dropoff Analysis.ipynb](#)

3.2.2 Operational Effect of Contractor Reliance

Variable	Correlation	Interpretation
B2B	-0.34	Negative correlation
End Customer	+0.42	Moderate positive correlation
Closing Stock	+0.09	Barely a relationship

Excess Stock	-0.28	Moderate negative
Total Sales	+0.43	Strongest positive correlation

Table 3.2.2: Correlation of Contractor Sales with Key Variables and Interpretation

Contractor sales show a weak negative correlation with B2B sales (-0.34), hinting at possible resource competition despite B2B's high AOV. A moderate positive link with End Customer sales (0.42) suggests seasonal complementarity (e.g., Oct-24 peaks). The negative correlation with excess stock (-0.28) implies possible inventory clearance during high contractor activity, though overstock issues persist. Minimal correlation with closing stock (0.09) and moderate alignment with total sales (0.43) highlight the contractors' volume-driven nature. However, these correlations do not imply causation and may be confounded by seasonality, policies, or data limitations, especially given proportional sales estimates and reliance on A-list months.

3.3 Volatility Diagnostics Results for Demand Stability

Findings from CV, MoM MAD, and ACF Plot

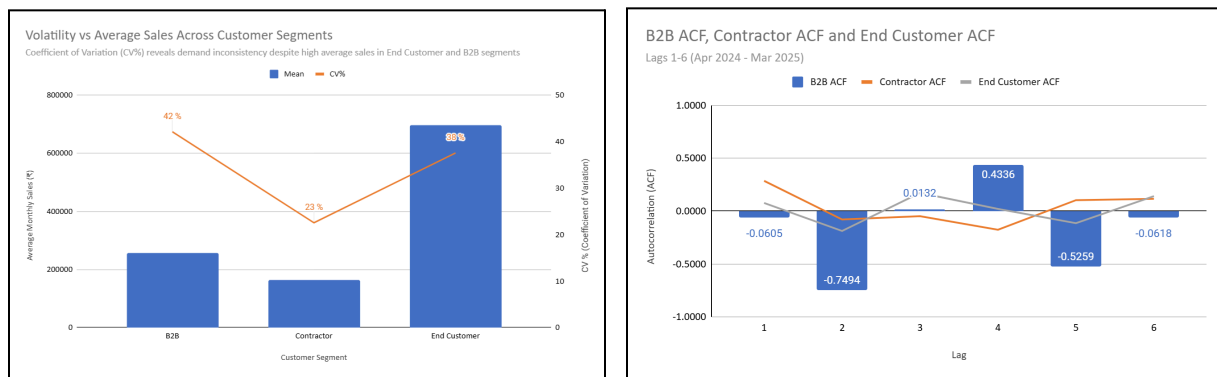


Figure 3.3 [a] & [b]: Combo charts for Volatility vs Average Sales and ACF plot Across Customer Segments

B2B (CV 42%) sees demand swing $\pm\text{€}42$ for every $\text{€}100$ sold, making it the most volatile relative to scale. **End Customers (CV 38%)** show large absolute fluctuations (MAD $\text{€}2.76\text{L}$) but are proportionally more stable. **Contractors (CV 23%, MAD $\text{€}32\text{K}$)** remain the most consistent. While End Customers lead in total sales, their erratic monthly demand complicates forecasting. B2B, though smaller, is similarly unreliable. In contrast, Contractors offer the most predictable and stable demand profile, ideal for operational planning.

The End Customer segment, despite being known for seasonal spikes—especially around the Diwali period—did not exhibit statistically strong autocorrelation at relevant lags. This absence of detectable seasonality is not contradictory, but rather a limitation of the dataset. ACF is effective at capturing recurring patterns across multiple periods. However, with only one annual cycle available, the analysis lacks the temporal depth needed to surface effects like festive-season demand. Domain insights and sales data continue to support the existence of such seasonal surges. Thus, while ACF did not confirm a repeating pattern, it does not invalidate the presence of seasonal volatility, especially in the End Customer segment. Instead, it highlights the need for longer-term data to strengthen statistical evidence.

3.4 Seasonality-based EOQ Modeling and Stress Testing Using What-if Analysis

3.4.1 Observing Inventory KPIs

From May to September FY25, procurement failed to adjust to falling sales, leading to bloated inventory, peak holding costs (~₹24K/month), and turnover dropping below 40%.

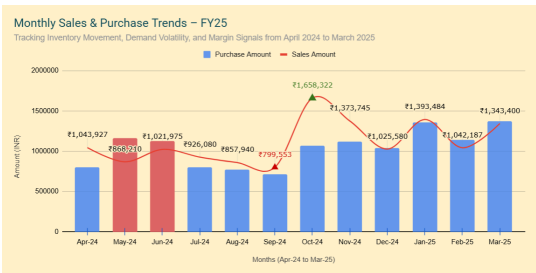


Figure 3.4.1: FY25 Inventory Dynamics: Sales vs. Purchase Trends (Apr 2024 – Mar 2025)

Excess stock peaked at 148.5% in September, with persistent double-digit wastage. In contrast, October to January showed high efficiency—timely procurement matched demand, turnover exceeded 80%, and margins improved (e.g., ₹6.36L profit in Oct). However, March saw a relapse, as purchases surged despite modest sales, pushing excess stock to

97.25%. These trends underscore the need for dynamic, season-aware procurement and real-time demand alignment.

3.4.2 Seasonality-Driven EOQ Modelling, System Benchmarking, and Annual Cost Simulation

The EOQ analysis revealed distinct optimal ordering strategies based on sourcing group and demand intensity:

Sourcing Group (Order Cost%)	Low Demand Months	Mid-Demand Months	High Demand Months	Avg Month	Steady State
Near-off (1.75%)	189	220	242	211	733
Far-off (2.75%)	224	276	303	265	918

Table 3.2.3 [a]: Recommendations by Sourcing Group and Demand Level (Units per Order)

Average order costs (1.75% for near-off, 2.75% for far-off) are midpoints of their respective ranges, applied to an average order value of ₹15,048.56. Under steady-state demand, optimal EOQs are 733 units (near-off) and 919 units (far-off), serving as baseline estimates. In high-demand months, demand-based EOQs drop to ~242 (near-off) and ~303 units (far-off), reflecting the need for larger, less frequent orders from distant sources due to higher logistics costs. While EOQ-optimized frequencies suggest 15–20 orders per month, the current system over-orders with 65–74 monthly orders, indicating significant inefficiency.

Demand Type	Current Order Freq	EOQ-based Order Freq (Near-off)	EOQ-based Order Freq (Far-off)
Low Demand	65.6	17.7	14.1
Mid Demand	70.8	20.6	16.5
High Demand	73.5	22.8	18.1
Avg Month	69.1	19.8	15.9
Annual Orders	829	68.8 (steady demand assumption)	54.9 (steady demand assumption)

Table 3.2.3 [b]: Comparison of Current vs. EOQ-Based Ordering Frequencies Across Demand Types

This indicates that current procurement is highly fragmented, resulting in avoidable transaction costs and logistical strain. Under a steady-state EOQ model, annual order count could be **reduced from 829 to approximately 54.9 (Far-off sourcing) or 68.8 (Near-off sourcing)**, implying cost savings and process streamlining opportunities

The optimized EOQ system—factoring in 1.75% (Near-off) and 2.75% (Far-off) order costs—suggests increasing order sizes to 179–287 units from the current 60, based on 50,413 units across 829 orders. This reduces monthly order frequency from 65.6–73.5 to 14.14–22.78, aligning better with seasonal demand (e.g., 3,011.5 units in low-demand months vs. 4,907.5 in peak months). While the current frequent-small-order strategy limits holding costs, it drives up ordering costs and causes inefficiencies, like a 148.51% overstock in Sep-24. The optimized

model cuts ordering costs (₹263.35–₹413.83), improves inventory turnover, and manages seasonal peaks more efficiently, though it slightly raises annual holding costs (₹49.41/unit).

System	Orders/Year	Avg Order Size	S (₹)	Annual Ordering Cost
Current (Status Quo)	829	60.8	338.59	₹2,80,592
EOQ (Near-off)	68.8	733	263.35	₹18,123
EOQ (Far-off)	54.9	918	413.83	₹22,743

Table 3.2.3 [c]: Annual Ordering Cost Comparison – Current System vs. EOQ Models

The current system’s 829 orders per year are highly fragmented and inefficient. EOQ analysis suggests this could drop by over 90%—to around 68 annual orders—without compromising demand fulfillment. While real-world factors (supplier limits, storage, and batching) may prevent such a sharp reduction, the takeaway is clear: fewer, larger orders = leaner, cheaper operations.

3.2.4 What if Results

WHAT-IF		TEST ↓		TEST ↓
Location	Order % \ Holding %	15%	20%	25%
Near-off (Inside Aligarh)	1.50%	7837	678	607
	2.00%	904	783	700
Far-off (Outside Aligarh)	2.50%	1011	876	783
	3.00%	1108	959	858
TEST →	3.50%	1197	1036	927

Table 3.2.4: Sensitivity Analysis of Total Procurement Cost under Varying Order and Holding Cost Scenarios

Increasing order cost to 3.5% due to fuel price hikes. Raising holding cost to 25% if storage costs rise. The what-if analysis shows that the total procurement cost rises with both order and holding cost percentages. Near-off sourcing (within Aligarh) is cheapest under low-cost conditions—₹607 at 1.5% order and 15% holding cost. As these rates increase, costs rise sharply (e.g., ₹1,197 at 3.5% and 25%). Far-off sourcing (outside Aligarh) consistently incurs higher costs, making it less economical, especially under higher cost scenarios. This sensitivity analysis underscores the need to align sourcing decisions with prevailing cost structures.

3.5 Tracing Credit Risk Concentration and Liquidity Strain in Informal Payment Flows

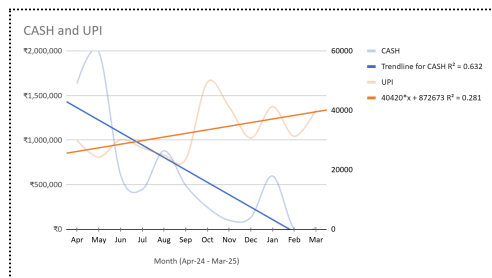


Figure 3.5: Dramatic Decline and Rise of Cash and UPI payments, respectively

With $R^2 = 0.632$, the downfall of cash transactions throughout the year has occurred faster than the rise of UPI payments. The analysis revealed that **UPI payments accounted for 98.45%** (₹1.31 Cr) of all sales, with contractors and end customers representing the majority. Within this, **₹31–₹41 lakh** were unregistered credit-based UPI sales, implying

₹5–7 lakh for contractors and **₹25–33 lakh** for end customers. These estimates align with the owner's rough figure of **₹16–18 lakh** in locked receivables (Table A: point[5])—around **5% of annual revenue** (Table A: point[8])—suggesting that a meaningful portion of cash flow is at risk. Assuming a 1-month credit term and that 35% of such credit sales remain unpaid, this translates to a **DSO (Days Sales Outstanding)** of ~30 days. In extreme cases (e.g., delays up to 2 years on ₹16–18 lakh), the business faces chronic liquidity strain (Table A: point [4]). Additionally, the **-0.2759 correlation between contractor sales and excess stock%** (Table 3.2.2) suggests that active contractor sales may relieve inventory pressure. However, this could also be coincidental, driven by **seasonal demand cycles or purchase behavior**, so causality cannot be definitively claimed.

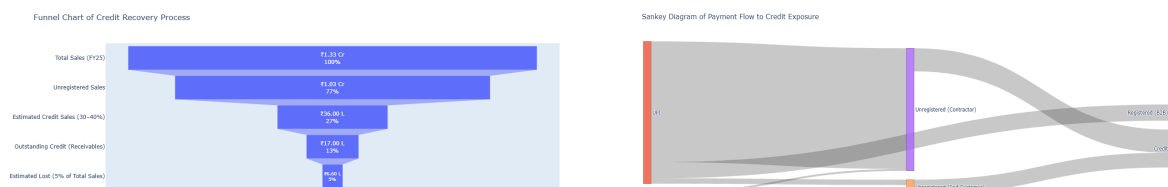


Figure 2.2 [a] & [b]: Quantifying Financial Leakage in the Credit Recovery Process and Tracing Financial Risk through Payment Flows

The Sankey diagram visualizes the flow of payments across buyer roles and highlights where credit risk originates. All registered B2B sales (₹30.6L) are routed through UPI and pose no credit risk, confirming the owner's claim of zero exposure in this segment.

In contrast, unregistered channels—Contractors and End Customers—receive the majority of UPI and Cash inflows and account for the entire ₹17L in outstanding receivables. Approximately 35% of their sales (₹3.6 Cr) are estimated to be on credit, validating the business owner’s estimate of ₹16–18L stuck in the market. This makes unregistered channels the sole contributors to credit risk, with End Customers forming a surprisingly large share of delayed payments. The diagram clarifies the operational stress introduced by these buyers and emphasizes the need for tighter credit controls and improved collection strategies in the unregistered segment.

4. Interpretation of Results and Recommendations

4.1 Interpretations of Results

4.1.1 Contractors Are High-Leverage Enablers, But Operationally Fragile

Contractors may only place small direct orders (₹1,000), but they consistently trigger large EC purchases (₹30,000+). Their indirect impact is substantial. However, the channel is fragile, overly reliant on a few top contributors. Losing just the top 20 contractors can erase over half of contractor-driven revenue, making proactive management non-negotiable.

4.1.2 The EC Segment Is the Strategic Profit Engine

Regardless of whether sales are organic or contractor-influenced, ECs deliver superior margins and order values. Statistical tests show no meaningful difference between subtypes, so the focus should be on safeguarding and scaling the EC base as a whole; it's the backbone of profitability.

4.1.3 Channel Mix: Balancing Risk and Reward

Contractor involvement boosts revenue but introduces volatility. Simulations show removing them shrinks revenue by ~4% but also stabilizes it. A purely organic model is safer, but less lucrative. The goal isn't to cut contractors, it's to formalize and de-risk their role within a balanced channel strategy.

4.1.4 Retaining Contractors is Costly, Losing Them is Costlier

Commission payouts of ₹1.2–1.9L per cycle are steep, but losing contractor-led revenue (~₹16–18L) is worse, especially with a 4-month recovery lag. A dual strategy is essential: retain key contractors while accelerating direct EC acquisition to reduce overdependence.

4.1.5 Contractor Impact is Systemic, Not Just Commercial

Contractors influence far more than revenue—they affect inventory turnover, credit exposure, and liquidity. Their role in clearing excess stock or delaying payments scales quickly and must be actively managed, not left to chance. They're not just intermediaries; they're operational levers.

4.1.7 Seasonal Blindness and the Case for EOQ Optimization

From May to September FY25, procurement failed to adjust to falling demand, leading to ballooning excess stock (up to 148%) and peak holding costs. While October–January showed improvement, March relapse proves the system lacks **seasonal awareness** and is overly **manual or reactive**. EOQ modeling shows stark inefficiencies: 829 orders per year, averaging just ~61 units, when optimal batch sizes are 733–918 (based on near/far sourcing). Even basic EOQ compliance could reduce ordering frequency by 90%, with **ordering cost drops from ₹2.8L to ₹18–22K/year**. Although these figures may present a hint of heavily assumed data, they surely have the potential directional cues for the business owner to move forward with.

4.1.7 Structural risks present solely in Unregistered Sales (Mostly ECs)

The risk here is the absence of formal credit terms, tracking, or recovery processes. Which creates an invisible leak that might slowly drain liquidity.

4.2 Recommendations for the Business Owner

Replace the current annual review process (during Diwali bonuses rollout) with **monthly reviews** to track churn risk and preempt disengagement. Initiate **formal Service Level Agreements (SLAs)** or **tier-based loyalty programs** for contractors like ABHISHEK, RAJKUMAR, BABLU SHARMA, DURGESH JI, and others in the top 10–20 individuals.

For Contractors, committing to faithfulness and waiting a whole year for bonuses while trying to meet sales targets requires a lot of motivation, if not trust. Competitors may attract such contractors, resulting in easy churn. Include performance-linked incentives such as **monthly bonuses, cashback, priority inventory, or co-branding offers**. Pilot with top 10 contractors in **60 days**, expand by **Q3 FY26**.

Focus on rebalancing the contractor lists by targeted onboarding from B/C-list contractors while keeping retention efforts to only A-List. Integrate **promo codes, cashback offers**, and

fast-delivery benefits on the official digital sales platform to attract organic customers. Educate organic customers by incentivizing them with similar offers to help grow trust and expand the organic customer channel within the End Customers channel. Use this **balanced channel strategy** that integrates contractors into a formal pipeline while simultaneously **building direct EC relationships**. Use channel-mix targets by aiming to first gradually lower the contractor-influenced EC sales to 50-60% by Q3 FY26 and set quarterly reviews to rebalance. Since contractors don't just affect sales, they influence operations. Observe contractor activity as a lead indicator for **inventory movement, credit stress, and demand planning**. **Maintain a digital ledger** or a sophisticated CRM add-on for tracking credit-based sales at the customer level. Implement **"soft lock" rules**: no further dispatches to customers with unpaid dues beyond a threshold. Introduce a **"Credit-Safe Seller" badge** for ECs completing 3-5+ timely payments.

4.3 Implementation Impact and Benefits

Implementing these recommendations will yield significant benefits. Urgent actions will reduce contractor churn risk (47.17% loss) by boosting Organic EC sales by ₹4.17 lakh and cutting credit exposure by ₹6-11 lakh, improving liquidity by ₹12.8 lakh during the 4-month recovery lag. Long-term diversification to B2B will add ₹19.4 lakh in stable revenue, leveraging its ₹29,160 AOV, while EOQ optimization will save ₹2.63 lakh annually in ordering costs, lowering holding costs and excess stock (e.g., from 148.51% to <50%). This holistic strategy will enhance channel diversity, strengthen direct relationships, and improve operational efficiency, positioning the business to withstand contractor attrition and seasonal volatility by July 2026.